

Dear Referee,

Thank you for carefully reading the previous version of the paper and providing me with great comments and suggestions. I think that the paper has improved substantially thanks to your suggested revisions.

This letter has two sections:

1. The first section summarizes the key changes to the manuscript.
2. The second section reports a detailed answer to each of the comments and suggestions in your letter. For each point, I first report your comment (in blue) and then discuss the paper modifications and extended analysis that address it.

Sincerely,

Lulu Wang

Summary of the Main Changes

This revision strengthens the reduced-form evidence, clarifies data limitations and modeling assumptions, estimates alternative specifications with debit-credit substitution, expands the counterfactual analysis, and rewrites the text for a broader audience.

1. **Reduced Form.** The Durbin section now emphasizes that interchange fees steer payment choices through multiple channels (rewards, advertising, teller incentives). New bank-level data show that among the banks that paid debit rewards pre-Durbin, those that ended rewards saw debit card volumes grow more slowly than those that kept rewards by a margin similar to the baseline difference-in-difference estimate. I also clarify the selectedness of the merchant event studies and expand the Discover analysis with a formal fixed-effects specification.
2. **Modeling.** A new “Key Assumptions” discussion clarifies what the data can and cannot identify. Merchant-type estimation is framed as a calibration anchored by the grocery-chain event studies. Fixed costs of acceptance cannot be separately identified. The counterfactuals are short-run predictions that hold non-rewards characteristics fixed.
3. **Credit-Debit Substitution.** The baseline keeps credit and debit segmented at the point of sale. Antitrust testimony and consumer surveys support segmentation, but Discover’s rewards experiment shows some substitution at the margin. Appendix OF estimates two specifications that relax segmentation; welfare results are close to the baseline for every counterfactual other than uncapping debit.
4. **Counterfactuals.** The main fee cap is now 120 bps, half-way between the previous cap and current levels. The relationship between cap strictness and welfare is concave, so this cap captures most of the welfare gains. New analyses decompose welfare gains into channels and model a dual-routing mandate. Re-estimating with income-dependent credit access yields similar results (\$31 bn vs. \$27 bn).
5. **Exposition.** The introduction is rewritten for a broader audience, built around standard IO concepts with the Durbin Amendment introduced in the opening paragraphs. Only three appendices are primary, with supplementary material in a separate Online Appendix. Footnotes and institutional digressions are shortened throughout.

Detailed Response to Referee 4

Below, I group together three related comments about the substitution between debit and credit.

The interpretation of the Durbin regulation is problematic. The 29% decline in signature debit volume for regulated issuers is taken as evidence of reward sensitivity, but debit use continued to grow overall, largely replacing cash, while exempt issuers promoted debit more aggressively post-Durbin. The difference-in-difference estimate does not imply reward-driven substitution to credit, and the claim of high reward sensitivity is overstated.

The claim that Durbin reduced welfare by shifting debit users to credit is likewise speculative. Debit and credit cards are not interchangeable for many consumers—credit constraints, budgeting preferences, or maxed-out credit lines prevent substitution. The paper argues that debit and credit may be substitutes at the adoption stage, but this is only a theoretical possibility, without empirical backing, and no existing studies have documented substantial migration from debit to credit following Durbin.

Without a convincing interpretation, the model likely misestimates reward sensitivity and debit-credit substitutability, risking misspecification and flawed policy conclusions. If debit and credit are not substitutes, Durbin may have improved welfare.

Reply: You are right that cleanly identifying debit-to-credit substitution is difficult, and no single piece of evidence is dispositive. The core question is whether credit constraints, budgeting preferences, or maxed-out credit lines prevent debit users from substituting toward credit.

I am not the first paper to suggest that Durbin caused migration from debit to credit. Mukharlyamov and Sarin (2025) exploit geographic variation in Durbin exposure and find that more-affected areas saw larger increases in credit card usage, providing direct evidence of debit-to-credit migration at the margin.

Two independent datasets suggest that many consumers choose debit as an active choice and not because they are constrained. In the NielsenIQ Homescan panel, around 75% of primary debit card holders also use a credit card (Figure 6). In the DCPC, Table 1 shows that 79% of debit-primary consumers hold a credit card, 53% hold a rewards credit card, and their average credit utilization is only 27%. These consumers have the option of paying with credit but choose debit, so access is not the key barrier.

Even among the minority who do not hold a credit card, marketing surveys suggest that the inability to qualify is not the primary barrier. Around 37% cite debt aversion (“prefer

not to carry any debt”), versus only 26% who report not qualifying (Boehm 2018). In the second-choice survey (Table OD.4), 47% of debit users report they would switch to credit if debit were unavailable.

Even if constraints matter, they do not affect the estimated welfare results. Online Appendix OG.2 re-estimates the model with binding credit constraints calibrated from credit score data. The counterfactual welfare effects are similar to baseline, because the consumers on the debit-credit margin are predominantly those with available credit capacity.

I agree that debit card volumes have grown dramatically over the past 10 years, but that too is consistent with my model. Aggregate trend data are descriptive only, but the pattern is consistent with substitution. Figure A.2 shows that debit growth slowed relative to its pre-Durbin trajectory while credit exceeded its trend. Post-Durbin volumes confound the fee cap with concurrent changes—dual routing mandates, the premium credit card expansion (e.g., Chase Sapphire Reserve in 2016), and AmEx’s merchant fee cuts—which is why the within-issuer difference-in-differences is the preferred identification strategy.

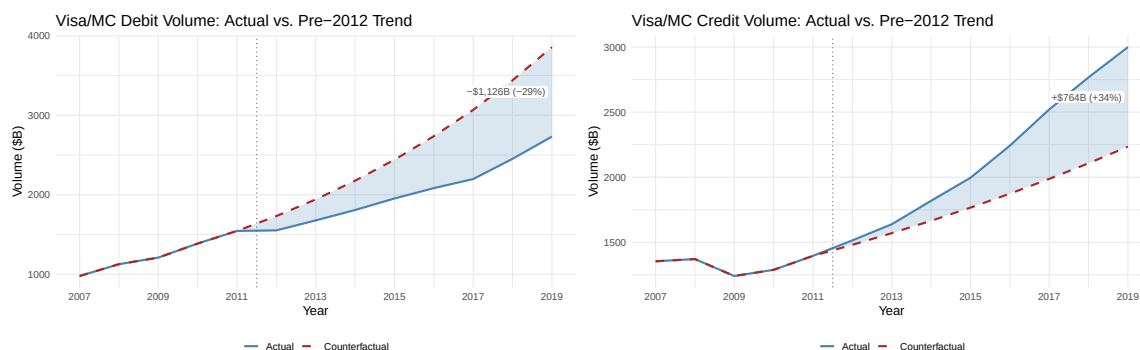


Figure A.2: Aggregate debit and credit card volumes over time, with pre-Durbin trend extrapolations

Notes: Pre-Durbin trends are linear trends fit to 2007–2011 log volumes and extrapolated forward.

Australia provides a rough external plausibility check on the magnitude of migration between debit and credit under tighter regulation. Australia caps both credit and debit interchange and has a debit:credit ratio of roughly 59:41 (Caddy et al. 2020), compared with 45:55 in the United States under current regulation. The closest analog in the model is the tourist test counterfactual (Online Appendix OG.4), which caps both credit and debit at the cost of cash. It implies a card-only debit:credit ratio of roughly 69:31, somewhat more debit-heavy than Australia. The magnitudes are in the same ballpark, and the small remaining gap is in the direction the model predicts: the tourist test caps

credit more tightly than Australia does, which pushes the mix further toward debit.

As a direct test of whether the estimated reward sensitivity is overstated, Online Appendix OG.3 halves the targeted Durbin moment, so the model matches only half the observed decline in debit volumes. This reduces reward sensitivity from 6.75 to 5.92, but the halved moment produces negative estimated marginal costs for Visa, which are difficult to reconcile with accounting data. The qualitative welfare conclusions are nonetheless preserved: capping merchant fees still improves welfare, and uncapping the Durbin Amendment would also improve welfare. Reducing reward sensitivity makes the merger-to-monopoly counterfactual harmful because networks have more market power over consumers.

Before proceeding to quantitative counterfactuals, the paper should establish key model properties. Define a socially optimal benchmark and illustrate divergences from it. This would clarify mechanisms and improve interpretation.

Reply: The theoretical literature provides the relevant benchmark in simpler settings. Rochet and Tirole (2011) show that the welfare-maximizing interchange fee equates the merchant fee to the cost of cash when networks lack market power. Edelman and Wright (2015) show that intermediation reduces welfare when platforms extract surplus through price coherence. Both results rest on the same mechanism: consumers do not internalize the merchant fee's effect on retail prices. That mechanism is present in my model, so the cost-of-cash cap retains its interpretation as the social benchmark.

The cost-of-cash cap (approximately 30 bp) replicates the Rochet-Tirole tourist test, and Online Appendix OG.4 confirms that even the more moderate 120 bps cap captures roughly 77% of what the Ramsey planner's solution delivers. The contribution of the paper is not to characterize the optimum but to quantify the welfare effects in a setting with realistic consumer heterogeneity, multi-homing, and network competition.

The counterfactuals are incomplete. The paper shows capping both debit and credit card fees helps, and that capping debit alone hurts. But what about capping credit card fees only? What is the optimal cap level? These questions remain unanswered.

Reply: The revised draft adopts this suggestion: the main counterfactual caps credit card merchant fees at 120 bp while leaving the Durbin Amendment cap on debit unchanged. Together with the appendix analysis, the paper now covers the full space of fee cap

policies. The credit-only cap at 120 bps generates \$27 billion in welfare gains (Section VI). The tourist test in Online Appendix OG.4 caps both credit and debit fees at the cost of cash (roughly 30 bps), adding only \$2 billion more. The Ramsey planner provides the optimal cap level computationally: it achieves \$35 billion in total gains, so the 120 bps credit-only cap already captures roughly 77% of the planner's welfare gains without directly regulating rewards.

The finding that competition reduces welfare is based on an extreme case (merging all networks into one). Yet the paper also shows that adding a public debit network helps. What is the optimal number and composition of networks? Please explore or discuss.

Reply: The model is not well-suited to determine the optimal number of networks. Introducing a new network requires specifying its unobserved characteristics and generates new logit shocks that mechanically affect adoption patterns. Without a strong microfoundation for what these additional characteristics represent, the optimal network count would hinge on assumptions that the model cannot discipline.

The public debit counterfactual is not intended as a statement about optimal network composition. It addresses a specific policy question—whether expanding public presence in payment markets, as several countries have considered, would improve welfare.

The assumption that debit and credit card features are fixed and unaffected by fees is vulnerable to the Lucas critique. Post-Durbin, banks changed checking account terms (fees, minimum balances). Similarly, if credit fees were capped, issuers could tighten criteria or change service quality. Without accounting for issuer responses, the welfare analysis is incomplete.

Reply: You raise a valid concern. The model is best interpreted as providing short-run predictions in which card characteristics are held fixed. In the longer run, issuers could respond to fee regulation by adjusting credit criteria, reducing non-pecuniary rewards or service quality in ways the model does not capture.

The Australian experience provides some reassurance on this point. Australia has had interchange fee caps since 2003, and Appendix Figure OB.1 shows that non-rewards annual fees did not rise, and the spread between credit card rates and term loans was unchanged following regulation. However, I cannot rule out that issuers adjusted non-price dimensions of quality. I have added language in Section IV.F acknowledging this

limitation and noting that the welfare counterfactuals are most reliable as short-run predictions. Section VI.F contextualizes this limitation alongside other sensitivity analyses in the main text. To the extent that issuers would respond to fee caps by reducing card quality, the model would overstate consumer welfare gains from regulation.

References

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